WEEK 6 PROJECT PROGRESS REPORT

PRJ-18: Real-Time University Bus Tracking System

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Project Guide:

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Reporting period: Week 6 | 18 October 2026 test evidence

https://github.com/BirendraPrasad/University-Bus-Tracking

Week 6 at a Glance

- Verified: trip scheduling API returned HTTP 200 for Trip ID 5.
- Verified: driver dashboard displayed the tested trip during the test.
- Verified: browser GPS submissions were stored through Kafka into PostgreSQL.
- Pending: student API latest GPS response and automatic map marker update.
- Pending: ETA, Admin Dashboard frontend, and full end-to-end regression tests.

1. Introduction

This standalone report records Week 6 implementation and testing progress for PRJ-18, the Real-Time University Bus Tracking System. It is intended to be submitted separately from the main project documentation.

Week 6 focused on validating trip scheduling, driver-side trip visibility, and the GPS ingestion pipeline from the browser through FastAPI and Kafka into PostgreSQL. Student-side map movement and ETA are not reported as completed because end-to-end verification is still pending.

2. Week 6 Objectives

- Validate trip creation through the backend API.
- Confirm that a scheduled trip is visible in the Driver Dashboard during testing.
- Verify that browser-generated GPS coordinates are sent to the server.
- Confirm GPS records are persisted in PostgreSQL through the Kafka consumer pipeline.
- Identify remaining work for student-side live tracking and end-to-end validation.

3. System Components and GPS Data Flow

The project uses FastAPI for backend API handling, PostgreSQL for persistent storage, Kafka for location-message transport, a GPS consumer to process messages, and browser-based frontend pages.

Driver browser GPS	FastAPI backend	Kafka topic: bus-location	GPS consumer	PostgreSQL: bus_locations

The diagram describes the tested GPS-ingestion path. It does not imply that the Student Dashboard map automatically moves in response to each location update.

4. Trip Scheduling Test

The POST /trips/ endpoint was tested through Swagger. The request returned HTTP 200 and created Trip ID 5.

Field	Observed value
Trip ID	5
Bus ID	1
Route ID	1
Driver ID	1
Trip date	18 October 2026
Initial status	SCHEDULED
Observed API result	HTTP 200 — Trip scheduled successfully

During subsequent dashboard testing, Trip 5 appeared under available/current trip information. This verifies the tested trip-scheduling path for this case, not every possible trip status transition.

5. Driver Dashboard and GPS Submission

During Week 6 testing, the Driver Dashboard displayed the tested trip and browser GPS coordinates were sent to the server. The interface reported that the location was sent. A database check provided evidence that GPS records were persisted.

- Driver-side trip visibility: observed for Trip ID 5.
- Browser GPS submission: tested.
- Persistence of submitted location records: verified in PostgreSQL.
- Physical vehicle movement and continuous student-side map movement: not verified by this test.

6. GPS Database Evidence

A query against the bus_locations table returned six records associated with Trip ID 5. The latest observed record in that test was:

Field	Observed value
Trip ID	5
Latitude	13.2407685
Longitude	77.7174273
Timestamp	2026-10-18 12:42:31.110847
Speed	NULL

These values are a historical test observation from 18 October 2026, not a claim about the bus current location. The speed field was NULL in this observed record.

7. Test Results

Test ID	Test	Observed result	Status
W6-01	FastAPI backend startup	Backend ran successfully during development.	PASS (observed)
W6-02	Create trip via POST /trips/	HTTP 200; Trip ID 5 created.	PASS
W6-03	Driver Dashboard trip display	Trip 5 appeared in available/current trip UI during testing.	PASS (observed)
W6-04	Browser GPS submission	Coordinates were sent to the server.	PASS (observed)
W6-05	GPS persistence	Six records were found for Trip 5 in bus_locations.	PASS (observed)
W6-06	Latest GPS in student API	Inspected endpoint did not include latest GPS in observed query/response implementation.	PENDING
W6-07	Automatic student map marker update	End-to-end behavior not verified.	NOT TESTED
W6-08	ETA display	ETA feature not verified as implemented.	PENDING
W6-09	Admin Dashboard frontend	Admin UI not present in inspected frontend file set.	PENDING

PASS is used only for behavior observed during the Week 6 test. Pending and untested cases are not represented as successful.

8. Student Live Tracking: Current State

The student endpoint GET /students/me/routes/{route_id}/bus was inspected. The observed query did not include the latest bus_locations GPS record. Therefore, the complete driver-to-student live-tracking path remains unfinished.

- Add a latest-location lookup for the relevant active trip to the student API response.
- Return latitude, longitude, location update timestamp, and speed when available.

- Update the Student Dashboard Leaflet map to display the bus marker.
- Refresh the marker as new location records arrive.
- Test the full workflow using a driver GPS update and verify the corresponding marker change on the student map.

9. Limitations and Pending Work

Priority	Remaining task	Completion evidence required
High	Student API latest GPS integration	API response includes latest coordinates for the selected route/trip.
High	Student map marker refresh	A new driver GPS update visibly updates the student map marker.
High	End-to-end tracking test	Documented test from driver browser to student map.
Medium	ETA calculation and display	ETA is calculated and shown with a defined method and test evidence.
Medium	Admin Dashboard UI	Admin pages implemented and connected to existing APIs.
Medium	Regression testing	Core login, trip, GPS, and role-protected API cases rerun.

Timetable and notification features should remain pending or not verified unless separate implementation evidence confirms completion.

10. Week 7 Action Plan

1. Update the student bus API to include the latest GPS record for the selected active trip.
2. Connect the student map marker to the returned latitude and longitude.
3. Implement periodic refresh or another suitable update mechanism and handle missing or stale location data.
4. Run an end-to-end test: create/activate a trip, send GPS from the driver page, verify database insertion, and confirm student map update.
5. Begin the Admin Dashboard frontend using existing APIs.
6. Document test results with genuine screenshots and SQL/API evidence.
7. Proceed to ETA after the location-update path is reliable.

11. Conclusion

Week 6 validated important parts of the backend GPS ingestion workflow: trip creation through the API, driver-side trip visibility, browser GPS submission, Kafka-based processing, and persistence of GPS records in PostgreSQL. The observed database contained six location records for Trip ID 5.

The project is not yet verified as a complete end-to-end student live-tracking system. Student API latest-location integration, automatic map marker movement, ETA, Admin Dashboard frontend, and broader regression testing remain pending. These items are the main priorities for the next development week.

12. Revision Record

Version	Scope	Notes
Week 6 — Separate Report	Implementation and validation update	Records observed test results, GPS pipeline evidence, limitations, and next steps. Intended to accompany the main project report.